AI Paper Evaluation Report

Student Details

Name : Ashwin Kirbaa

Register No : 25cse028

Department : CSE

Semester : 3

Teacher Details

Teacher : Dexter Morgan

Subject : TwinEval

Overall Result

Total Marks	35.45 / 50
Percentage	70.9%
Grade	B+
Result	Pass

Overall Strengths

• Correct answer.

• Correctly identified key concept(s): Autoencoder Architecture.

• Clear structure and formatting presented.

• Correctly identified key concept(s): Generative Models, Adversarial Training, Discriminator, Generator.

• Demonstrated foundational understanding of core concepts.

• Correctly identified key concept(s): recurrent neural networks, gated recurrent units, hidden state, update gate, reset gate.

Overall Weaknesses

• Correct answer: binary.

• Correct answer: False.

• Correct answer: input and output gates.

• Correct answer: LSTM.

• Correct answer: context.

• Include missing concept: Reconstruction Error

Frequently Missed Concepts

• binary

• False

• input and output gates

• LSTM

• context

• Reconstruction Error

• Sparsity Constraint

• Feature Disentanglement

Topics to Revise

• A Deep Boltzmann Machine is a generative model composed of multiple layers of _

• Say True or False: Standard autoencoders always learn a probabilistic latent sp

• GRU architecture simplifies LSTM by merging the _____ and input gates.

• _____ generation models use LSTM networks to produce text responses in a con

• In opinion mining, recurrent architectures help to retain _____ across word

• Interpret the impact of attention mechanisms in improving the performance of im

Personalized Recommendations

• Revise these frequently missed concepts: binary, False, input and output gates, LSTM, context.

• Focus additional study time on the lowest-scoring questions/topics.

Study Suggestions

• Review and practice questions related to: binary

• Review and practice questions related to: False

• Review and practice questions related to: input and output gates

• Review and practice questions related to: LSTM

• Review and practice questions related to: context

Question-wise Evaluation

Q1. _____ network is designed to reduce overfitting by penalizing the Jacobian of the encoder output. a) Sparse Autoencoder b) Denoising Autoencoder c) Contractive Autoencoder d) VAE

Marks: **0.5 / 0.5** Percentage: 100% Grade: A+ Bloom Level: Remember Confidence: 95%

Feedback: Correct.

Found Concepts:

• c) Contractive Autoencoder

Missing Concepts:

None

Missing Keywords:

None

Missing Points:

None

Missing Diagrams:

None

Strengths:

• Correct answer.

Areas to Improve:

None

Suggestions:

None

Q2. A Deep Boltzmann Machine is a generative model composed of multiple layers of _____ variables.

Marks: **0 / 0.5** Percentage: 0% Grade: F Bloom Level: Remember Confidence: 95%

Feedback: Incorrect. Correct answer: binary.

Found Concepts:

None

Missing Concepts:

• binary

Missing Keywords:

• binary

• variables

Missing Points:

None

Missing Diagrams:

None

Strengths:

None

Areas to Improve:

• Correct answer: binary.

Suggestions:

None

Q3. Say True or False: Standard autoencoders always learn a probabilistic latent space.

Marks: **0 / 0.5** Percentage: 0% Grade: F Bloom Level: Remember Confidence: 95%

Feedback: Incorrect. Correct answer: False.

Found Concepts:

None

Missing Concepts:

• False

Missing Keywords:

• autoencoder

• latent space

• probabilistic

Missing Points:

None

Missing Diagrams:

None

Strengths:

None

Areas to Improve:

• Correct answer: False.

Suggestions:

None

Q4. The encoder in a sequence-to-sequence model is responsible for: a) Predicting output directly b) Mapping output to input c) Summarizing input sequence d) Generating noise

Marks: **0.5 / 0.5** Percentage: 100% Grade: A+ Bloom Level: Remember Confidence: 95%

Feedback: Correct.

Found Concepts:

• b) Mapping output to input

Missing Concepts:

None

Missing Keywords:

None

Missing Points:

None

Missing Diagrams:

None

Strengths:

• Correct answer.

Areas to Improve:

None

Suggestions:

None

Q5. Say True or False: Adversarial training involves only supervised learning principles.

Marks: **0.5 / 0.5** Percentage: 100% Grade: A+ Bloom Level: Remember Confidence: 95%

Feedback: Correct.

Found Concepts:

• False

Missing Concepts:

None

Missing Keywords:

None

Missing Points:

None

Missing Diagrams:

None

Strengths:

• Correct answer.

Areas to Improve:

None

Suggestions:

None

Q6. GRU architecture simplifies LSTM by merging the _____ and input gates.

Marks: **0 / 0.5** Percentage: 0% Grade: F Bloom Level: Remember Confidence: 95%

Feedback: Incorrect. Correct answer: input and output gates.

Found Concepts:

None

Missing Concepts:

• input and output gates

Missing Keywords:

• input gate

• output gate

Missing Points:

None

Missing Diagrams:

None

Strengths:

None

Areas to Improve:

• Correct answer: input and output gates.

Suggestions:

None

Q7. _____ technique is used for separating an image into distinct regions based on object boundaries. a) Captioning b) Segmentation c) Detection d) Translation

Marks: **0.5 / 0.5** Percentage: 100% Grade: A+ Bloom Level: Remember Confidence: 95%

Feedback: Correct.

Found Concepts:

• Segmentation

Missing Concepts:

None

Missing Keywords:

None

Missing Points:

None

Missing Diagrams:

None

Strengths:

• Correct answer.

Areas to Improve:

None

Suggestions:

None

Q8. Say True or False: Recursive Neural Networks are typically used for sentiment analysis at the sentence and phrase level.

Marks: **0.5 / 0.5** Percentage: 100% Grade: A+ Bloom Level: Remember Confidence: 95%

Feedback: Correct.

Found Concepts:

• True

Missing Concepts:

None

Missing Keywords:

None

Missing Points:

None

Missing Diagrams:

None

Strengths:

• Correct answer.

Areas to Improve:

None

Suggestions:

None

Q9. _____ generation models use LSTM networks to produce text responses in a conversation.

Marks: **0 / 0.5** Percentage: 0% Grade: F Bloom Level: Remember Confidence: 95%

Feedback: Incorrect. Correct answer: LSTM.

Found Concepts:

None

Missing Concepts:

• LSTM

Missing Keywords:

• LSTM

• generation models

• text responses

Missing Points:

None

Missing Diagrams:

None

Strengths:

None

Areas to Improve:

• Correct answer: LSTM.

Suggestions:

None

Q10. _____ task uses both CNN and LSTM together to generate textual context from visuals. a) Video classification b)

Image captioning c) Object tracking d) Opinion mining

Marks: **0.5 / 0.5** Percentage: 100% Grade: A+ Bloom Level: Remember Confidence: 95%

Feedback: Correct.

Found Concepts:

• b) Image captioning

Missing Concepts:

None

Missing Keywords:

None

Missing Points:

None

Missing Diagrams:

None

Strengths:

• Correct answer.

Areas to Improve:

None

Suggestions:

None

Q11. Say True or False: Convolutional Neural Networks are used for Named Entity Recognition tasks.

Marks: **0.5 / 0.5** Percentage: 100% Grade: A+ Bloom Level: Remember Confidence: 95%

Feedback: Correct.

Found Concepts:

• True

Missing Concepts:

None

Missing Keywords:

None

Missing Points:

None

Missing Diagrams:

None

Strengths:

• Correct answer.

Areas to Improve:

None

Suggestions:

None

Q12. In opinion mining, recurrent architectures help to retain _____ across word sequences.

Marks: **0 / 0.5** Percentage: 0% Grade: F Bloom Level: Remember Confidence: 95%

Feedback: Incorrect. Correct answer: context.

Found Concepts:

None

Missing Concepts:

• context

Missing Keywords:

• context

• word sequences

• recurrent architectures

Missing Points:

None

Missing Diagrams:

None

Strengths:

None

Areas to Improve:

• Correct answer: context.

Suggestions:

None

Q13. Compare the performance of standard Autoencoders vs. Sparse Autoencoders on high-dimensional data.

Marks: **1.64 / 2** Percentage: 82% Grade: A Bloom Level: Create Confidence: 96%

Feedback: Very good answer with strong coverage of the topic. The student demonstrates a basic understanding of the difference between standard and sparse autoencoders, particularly regarding

their efficiency with high-dimensional data. However, the answer lacks depth in explaining the reasons for the performance difference and the concept of sparsity. The answer receives partial marks for accurately describing the basic operation of each type of autoencoder and identifying the key difference.

Found Concepts:

• Autoencoder Architecture

Missing Concepts:

• Reconstruction Error

• Sparsity Constraint

• Feature Disentanglement

Missing Keywords:

• Dimensionality Reduction

• Sparsity

Missing Points:

• Sparse Autoencoders mitigate overfitting through the sparsity constraint.

Missing Diagrams:

• None

Strengths:

• Correctly identified key concept(s): Autoencoder Architecture.

• Clear structure and formatting presented.

Areas to Improve:

• Include missing concept: Reconstruction Error

• Include missing concept: Sparsity Constraint

• Include missing concept: Feature Disentanglement

• Elaborate further to cover all parts of the teacher's model answer key.

• Missing expected keyword(s): Dimensionality Reduction, Sparsity.

• Missing/incomplete diagram(s): None.

Suggestions:

• Study and review the missing concepts: Reconstruction Error, Sparsity Constraint, Feature Disentanglement.

• Excellent higher-order synthesis and problem-solving shown.

Q14. Determine the suitability of Generative Adversarial Networks (GANs) for real-world applications such as image enhancement and generation.

Marks: **1.67 / 2** Percentage: 83.5% Grade: A Bloom Level: Remember Confidence: 98%

Feedback: Very good answer with strong coverage of the topic. The answer demonstrates a basic understanding of GANs and their application to image enhancement and generation. However, it lacks a critical discussion of the training challenges inherent in GANs, and the explanation is somewhat simplified. The marks reflect this partial understanding and the inclusion of key concepts.

Found Concepts:

• Generative Models

• Adversarial Training

• Discriminator

• Generator

Missing Concepts:

None

Missing Keywords:

• GANs

Missing Points:

None

Missing Diagrams:

None

Strengths:

• Correctly identified key concept(s): Generative Models, Adversarial Training, Discriminator, Generator.

• Clear structure and formatting presented.

Areas to Improve:

• Expand on key concepts and definitions from the study material.

• Elaborate further to cover all parts of the teacher's model answer key.

• Missing expected keyword(s): GANs.

Suggestions:

• Explain the concepts in your own words rather than solely listing facts.

Q15. Interpret the impact of attention mechanisms in improving the performance of image captioning models.

Marks: **0.99 / 2** Percentage: 49.5% Grade: D Bloom Level: Apply Confidence: 94%

Feedback: Partial answer; several important concepts are missing. The answer demonstrates a basic understanding of CNN and LSTM, but fails to adequately explain the core concept of attention mechanisms and their impact on image captioning performance. The response is partially correct in identifying relevant components but lacks crucial details regarding the dynamic weighting of image features.

Found Concepts:

None

Missing Concepts:

• sequence-to-sequence learning

• visual attention

• neural networks

Missing Keywords:

• attention mechanism

• region-based

• context vector

Missing Points:

None

Missing Diagrams:

None

Strengths:

• Demonstrated foundational understanding of core concepts.

• Clear structure and formatting presented.

Areas to Improve:

• Include missing concept: sequence-to-sequence learning

• Include missing concept: visual attention

• Include missing concept: neural networks

• Elaborate further to cover all parts of the teacher's model answer key.

• Missing expected keyword(s): attention mechanism, region-based, context vector.

Suggestions:

• Study and review the missing concepts: sequence-to-sequence learning, visual attention, neural networks.

• Demonstrate practical implementation details in your answer.

Q16. Evaluate the suitability of GRU architecture for time-series forecasting by examining its structure and ability to model sequential dependencies.

Marks: **7.9 / 14** Percentage: 56.4% Grade: C Bloom Level: Remember Confidence: 92%

Feedback: Partial answer; several important concepts are missing. The student demonstrates a basic understanding of the GRU architecture and its components, but the explanation is incomplete and lacks depth. The student's application examples are incorrect, and the overall response needs more detail to fully address the question's requirements. The student's explanation of the equations is simplified, but generally correct.

Found Concepts:

• recurrent neural networks

• gated recurrent units

• hidden state

• update gate

• reset gate

Missing Concepts:

• temporal dependencies

Missing Keywords:

• gating mechanism

• sequential dependencies

• vanishing gradients

Missing Points:

• Describe how the gating mechanism allows GRUs to model sequential dependencies.

• Explain why GRUs are better than standard RNNs at handling long-range dependencies.

Missing Diagrams:

• Diagram of a GRU cell showing the input, hidden state, reset gate, update gate, and output would be beneficial (omitted for brevity).

Strengths:

• Correctly identified key concept(s): recurrent neural networks, gated recurrent units, hidden state, update gate, reset gate.

• Clear structure and formatting presented.

Areas to Improve:

• Include missing concept: temporal dependencies

• Elaborate further to cover all parts of the teacher's model answer key.

• Missing expected keyword(s): gating mechanism, sequential dependencies, vanishing gradients.

• Missing/incomplete diagram(s): Diagram of a GRU cell showing the input, hidden state, reset gate, update gate, and output would be beneficial (omitted for brevity)..

Suggestions:

• Study and review the missing concepts: temporal dependencies.

• Explain the concepts in your own words rather than solely listing facts.

Q17. Demonstrate the implementation of Named Entity Recognition (NER) using deep learning, and explain the steps involved in identifying entities in a text.

Marks: **11 / 14** Percentage: 78.6% Grade: B+ Bloom Level: Remember Confidence: 98%

Feedback: Very good answer with strong coverage of the topic. The student provides a reasonably complete explanation of NER, covering the key steps and techniques. While the explanation could be more detailed regarding the deep learning aspects and model architecture, the core understanding of the process and the BIO tagging method is clearly demonstrated. Deducting

marks is minimal due to the clarity of the explanation.

Found Concepts:

• Named

• Entity

• Recognition

Missing Concepts:

• Demonstrate

• implementation

Missing Keywords:

• Demonstrate

• implementation

• using

Missing Points:

None

Missing Diagrams:

None

Strengths:

• Correctly identified key concept(s): Named, Entity, Recognition.

• Clear structure and formatting presented.

Areas to Improve:

• Include missing concept: Demonstrate

• Include missing concept: implementation

• Elaborate further to cover all parts of the teacher's model answer key.

• Missing expected keyword(s): Demonstrate, implementation, using.

Suggestions:

• Study and review the missing concepts: Demonstrate, implementation.

• Explain the concepts in your own words rather than solely listing facts.

Q18. Apply an LSTM model to generate captions for video frames after preprocessing the video dataset.

Marks: **0 / 14** Percentage: 0% Grade: F Bloom Level: Remember Confidence: 100%

Feedback: No answer was provided for this question.

Found Concepts:

None

Missing Concepts:

• Apply

• model

• generate

• captions

• video

Missing Keywords:

• Apply

• model

• generate

• captions

• video

• frames

• after

• preprocessing

Missing Points:

None

Missing Diagrams:

None

Strengths:

None

Areas to Improve:

• No answer was provided for this question.

Suggestions:

• Attempt every question, even partially, for part marks.

Q19. Demonstrate the implementation of a Denoising Autoencoder for image reconstruction by outlining the key steps in training and testing the model with noisy image data.

Marks: **8.75 / 10** Percentage: 87.5% Grade: A Bloom Level: Understand Confidence: 92%

Feedback: Very good answer with strong coverage of the topic. The student demonstrates a basic understanding of the DAE concept and its function, correctly identifying the core steps. However, the answer is very high-level and lacks technical detail, particularly regarding the model architecture, training process, and evaluation metrics. The inclusion of the MSE formula shows a partial grasp of the underlying mathematical principles.

Found Concepts:

• autoencoders

Missing Concepts:

• representation learning

• non-linear dimensionality reduction

• feature extraction

Missing Keywords:

• neural network

• reconstruction

Missing Points:

• The process of adding noise to the training data is crucial for the DAE's learning process.

• The choice of loss function (MSE) is critical for guiding the training process.

Missing Diagrams:

• A simple block diagram illustrating the DAE architecture (encoder and decoder) would be beneficial.

Strengths:

• Correctly identified key concept(s): autoencoders.

• Clear structure and formatting presented.

Areas to Improve:

• Include missing concept: representation learning

• Include missing concept: non-linear dimensionality reduction

• Include missing concept: feature extraction

• Elaborate further to cover all parts of the teacher's model answer key.

• Missing expected keyword(s): neural network, reconstruction.

• Missing/incomplete diagram(s): A simple block diagram illustrating the DAE architecture (encoder and decoder) would be beneficial..

Suggestions:

• Study and review the missing concepts: representation learning, non-linear dimensionality reduction, feature extraction.

• Include real-world applications or examples to strengthen your explanation.

Q20. Develop a deep learning model using the VGG16 pre-trained architecture to classify images of fruits into three categories: Apple, Banana, and Orange.

Marks: **0 / 10** Percentage: 0% Grade: F Bloom Level: Remember Confidence: 100%

Feedback: No answer was provided for this question.

Found Concepts:

None

Missing Concepts:

• Develop
• learning
• model
• using
• VGG16

Missing Keywords:

• Develop
• learning
• model
• using
• VGG16
• pre-trained
• architecture
• classify

Missing Points:

None

Missing Diagrams:

None

Strengths:

None

Areas to Improve:

• No answer was provided for this question.

Suggestions:

• Attempt every question, even partially, for part marks.